

Syllabus

Course code: PH323

Course Name: Principal and Techniques in Molecular Biology

Pre-requisite: Essentials of Biochemistry

Objective of the course: To study about basic principles and techniques involved in molecular and genetic biology.

Course Content:

UNIT I: Basic principles of molecular biology (3L)

UNIT II: Genetic material – DNA and RNA– their structure and function, DNA Sequencing – Sanger vs Next Generation sequencing, recombinant DNA technology and Genetic analysis. (7L)

UNIT III: Prokaryotic and Eukaryotic gene structure and Function (3L)

UNIT IV: Replication, Transcription and Translation, Variations and gene mutations – Copy Number Variation, Single Nucleotide Polymorphism etc. Gene regulation – methylation and post-translation changes

UNIT V: Proteins– structure, purification, characterization and functional analysis (7L)

UNIT VI: Oncogenes and Tumor suppressor genes (3L)

UNIT VII: Genomes – Human vs bacterial/chloroplast/mitochondria vs Viral (3L)

UNIT VIII: Basics of techniques involve in molecular biology: Separation of DNA/RNA and protein centrifugation and gel electrophoresis, cell culture techniques, Gene Cloning and expression, Immunochemical techniques, Polymerase Chain Reaction, Microscopy and Chromatin Immunoprecipitation and ChIP-Seq (7L)

UNIT IX: Basics of computational techniques and Bioinformatics, Downloading a sequence of a “Specific” gene sequencing from a database, BLAST search, Primer design for the PCR amplification of a “Specific” gene/DNA sequence, Pair-wise alignment of two DNA sequences (6L)

Readings:

Text Books:

1. Leininger Principles of Biochemistry.
2. Biochemistry: by Jeremy M. Berg, John L. Tymoczko
3. HarIllustrated Biochemistry 30th Edition by Victor Rodwell, David Bender, P. Anthony Weil, Peter Kennelly, and Kathleen Botham